ADDING POINTS AND LINES TO GENERALIZED POLYGONS: THE CONSTRUCTIONS WITHOUT COORDINATES

DEGREE–DIAMETER LAB

Abstract. Written for Gabriela Araujo-Pardo and Ignacio López. Each construction of the lab is described here only in terms of incidence: points, lines, the lines through a point, the points of a line, flags, anti-flags, collinearity, distances in the incidence graph, and configurations (ovoids, spreads, m-ovoids, whiskers) defined by their combinatorial properties. For each construction we say what is chosen in the starting geometry, which incidences are deleted or moved, which new points and lines are added and exactly what they are incident with, which pairs of elements stop being collinear or concurrent and how the new elements repair this, and why the diameter and the degrees are as claimed. Section 1 treats the diameter-3 constructions in a projective plane: the three hats of Araujo-Pardo and López and our four hats, side by side and in the same words. Sections 2–4 treat diameters 4, 6 and 7 in generalized quadrangles and hexagons. The sections can be read independently.

Sources. Everything here is taken from the lab's notes: `research/theorems/POLISHED.md` (coordinates and complete proofs of all results below), `research/geometry/geometry.tex`, `research/plus2/coords/coords.tex`, `research/theorems/D7_HEXAGON.md` and the joint draft `research/paper/joint.tex`. No new result is claimed. Every combinatorial statement in Section 1 was checked by a program that builds both structures from the rules below on a plane given only by its incidence relation (`research/synthetic/checks/check_d3.py`, for $q = 3, 4, 5, 7$, for the non-Desarguesian nearfield plane of order 9, and against the coordinate version for $q \le 9$); the checks for the other sections are listed in Appendix A.

0. Conventions

An *incidence structure* consists of points, lines and a symmetric incidence relation I between points and lines. Points and lines are *elements*. A *flag* is an incident point–line pair, an *anti-flag* a non-incident one. The *degree* of an element is the number of elements incident with it. Two distinct points are *collinear* if some line contains both, and two distinct lines are *concurrent* if some point lies on both. We do *not* assume that two points lie on at most one line: after points and lines are added, two points may have several common lines, and two lines may have the same points.

The *incidence graph* has the elements as vertices and the flags as edges; it is bipartite. Distances $d(x, y)$ and the diameter are taken in it. So two points are at distance 2 when they are collinear, a point and a line are at distance 3 when they form an anti-flag and some point of the line is collinear with the point, and so on.

A *generalized n-gon* is an incidence structure whose incidence graph has diameter n and girth $2n$; elements at distance n are *opposite*. It has *order* (s, t) if every line has $s + 1$ points and every point is on $t + 1$ lines. Projective planes are the generalized 3-gons, generalized quadrangles the 4-gons, generalized hexagons the 6-gons. Write $n_b(\Delta, D)$ for the largest number of elements of an incidence structure with all degrees at most Δ and diameter at most D (the bipartite degree–diameter problem). A generalized n-gon of order (q, q) attains the counting (Moore) bound for $n_b(q + 1, n)$. For degree $q + 2$ there is no such object when $q + 1$ is not a prime power, and every construction below starts from a generalized polygon of order (q, q) (or $(q - 1, q + 1)$), and adds

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points and lines so that the maximum degree becomes $q+2$ (or more) while the diameter stays the same.

Distances can be read off from collinearity alone:

Lemma 0.1 (Diameter in incidence terms)**.** *Let $\mathcal{S}$ be an incidence structure in which every element is incident with at least one element.*

(a) $\operatorname{diam}\mathcal{S} \le 3$ *if and only if any two points are collinear and any two lines are concurrent.*
(b) $\operatorname{diam}\mathcal{S} \le 4$ *if and only if for every anti-flag (p, L) some point of L is collinear with p.*
(c) $\operatorname{diam}\mathcal{S} \le 6$ *if and only if for every anti-flag (p, L) some point of L is at distance at most 4 from p.*
(d) $\operatorname{diam}\mathcal{S} \le 7$ *if and only if any two points, and any two lines, are at distance at most 6.*

Proof. For an even bound D: two points $p \ne p'$ are at distance at most D as soon as p is within $D-1$ of some line L through p' (if $p \mathrel{\mathrm{I}} L$ the distance is 2); two lines likewise. For an odd bound D: an anti-flag (p, L) is within distance D as soon as p is within $D-1$ of some point of L. Distances between a point and a line are odd and distances between two points or two lines are even, which gives the converses. □

1. Diameter 3: hats in a projective plane

Let Π be a projective plane of order q: any two points are on exactly one line, any two lines meet in exactly one point, and every element has degree $q+1$. For points $u \ne v$ we write uv for the line joining them. Both constructions in this section keep all points and lines of Π, delete or move a few flags near a chosen anti-flag, and add a few points and lines. By Lemma 0.1(a) the diameter stays 3 exactly when any two points remain collinear and any two lines remain concurrent; this is what has to be checked.

1.1. **The frame.** Fix an anti-flag (A, ℓ). Every line through A meets ℓ in one point, and every point of ℓ is on one line through A. So the $q+1$ flags (X, AX), X a point of ℓ, are the only incidences between the points of ℓ and the lines through A, and they match the points of ℓ with the lines through A. Both constructions change only some of these $q+1$ flags.

Choose distinct *marked* points X_s on ℓ, indexed by a label set S, and put

$$\lambda_s = AX_s \qquad (s \in S).$$

One label $a \in S$ is distinguished, and $T = S \setminus \{a\}$. We write

$$\ell^* = \{\text{points of } \ell\} \setminus \{X_t : t \in T\}, \qquad \Lambda^* = \{\text{lines through } A\} \setminus \{\lambda_t : t \in T\}.$$

So ℓ^* contains X_a and the $q+1-|S|$ unmarked points of ℓ, and Λ^* contains λ_a and the $q+1-|S|$ lines joining A to them. We call A, ℓ, the X_s and the λ_s the *frame*.

- The construction of Araujo-Pardo and López [1] (*three hats*) uses three marked points: $S = \{0, 1, a\}$, $T = \{0, 1\}$.
- The *four hats* use four: $S = \{0, 1, a, c\}$, $T = \{0, 1, c\}$; this needs $q \ge 3$.

The labels $0, 1, a, c$ are only names. The new elements are, for each $s \in S$, a *hat point* H_s and a *hat line* K_s, and one more point Q and one more line M.

1.2. **The two constructions.** Both constructions follow the same six rules. They differ in two permutations of T: the *links* π, a permutation of T without fixed points, and the *rotation* ρ.

Definition 1.1. The incidence structure $\mathcal{S}$ has as points the points of Π, the point Q and the hat points H_s; as lines the lines of Π, the line M and the hat lines K_s; and exactly the following incidences.

(R1) The incidences of Π, except the flags (X_s, λ_s), $s \in S$; and the flags $(X_t, \lambda_{\rho(t)})$, $t \in T$.
(R2) H_s lies on every line of Π through X_s other than ℓ (q lines, λ_s among them).

(R3) K_s contains every point of λ_s other than A (q points, X_s among them).
(R4) M contains the points of ℓ^* and every hat point H_s.
(R5) Q lies on the lines of Λ^* and on every hat line K_s.
(R6) The *links*: $A \mathrel{\mathrm{I}} K_a$, $H_a \mathrel{\mathrm{I}} \ell$, and $H_t \mathrel{\mathrm{I}} K_{\pi(t)}$ for $t \in T$.

	three hats [1]	four hats
marked points	X_0, X_1, X_a	X_0, X_1, X_a, X_c
flags deleted	(X_a, λ_a)	(X_0,λ_0), (X_c,λ_c), (X_1,λ_1), (X_a,λ_a)
flags added	none	(X_0,λ_c), (X_c,λ_1), (X_1,λ_0)
H_s lies on	the q lines through X_s other than ℓ; M; one link line	
K_s contains	the q points of λ_s other than A; Q; one link point	
M contains	the $q-1$ points of ℓ other than X_0, X_1; H_0, H_1, H_a	the $q-2$ points of ℓ other than X_0, X_1, X_c; H_0, H_1, H_a, H_c
Q lies on	the $q-1$ lines through A other than λ_0, λ_1; K_0, K_1, K_a	the $q-2$ lines through A other than $\lambda_0, \lambda_1, \lambda_c$; K_0, K_1, K_a, K_c
links	$A \mathrel{\mathrm{I}} K_a$, $H_a \mathrel{\mathrm{I}} \ell$, $H_0 \mathrel{\mathrm{I}} K_1$, $H_1 \mathrel{\mathrm{I}} K_0$	$A \mathrel{\mathrm{I}} K_a$, $H_a \mathrel{\mathrm{I}} \ell$, $H_0 \mathrel{\mathrm{I}} K_c$, $H_c \mathrel{\mathrm{I}} K_1$, $H_1 \mathrel{\mathrm{I}} K_0$
π, ρ	$\pi = (0\ 1)$, $\rho = \mathrm{id}$	$\pi = \rho = \sigma = (0\ c\ 1)$: $0 \mapsto c \mapsto 1 \mapsto 0$
new elements	Q, H_0, H_1, H_a; M, K_0, K_1, K_a	Q, H_0, H_1, H_a, H_c; M, K_0, K_1, K_a, K_c

TABLE 1. The two constructions, rule by rule. With $\rho = \mathrm{id}$, rule (R1) deletes and restores the flags (X_t, λ_t), $t \in T$, so the three hats delete only (X_a, λ_a).

So the three hats of [1] are $\mathcal{S}$ with $S = \{0, 1, a\}$, $\pi = (0\ 1)$, $\rho = \mathrm{id}$, and the four hats are $\mathcal{S}$ with $S = \{0, 1, a, c\}$ and $\pi = \rho = \sigma$. Table 1 puts the two side by side, and Figure 1 shows what rule (R1) and the links do to the flags between ℓ and the lines through A.

Dictionary. In [1] the plane is described from a flag (P, L) (the point and line at infinity); P_0 is a point of L other than P, and L_0 a line through P other than L. In our names $P_0 = A$, $L_0 = \ell$, $(0, s) = X_s$, $[0, s] = \lambda_s$, $Q_0 = Q$, $M_0 = M$, $\widehat{(0,s)} = H_s$, $\widehat{[0,s]} = K_s$. By their Observation 1, the lines through $(0, s)$ other than L_0 are the $[t, s]$ and the points of $[0, s]$ other than P_0 are the (t, s), so their edges $(\widehat{(0,j)}, [t, j])$ and $(\widehat{[0,j]}, (t, j))$ are rules (R2) and (R3). Their edges (P, M_0) and $(M_0, (0, i))$, $i \neq 0, 1$, together say that M contains the points of L_0 other than $(0, 0), (0, 1)$, and with $(M_0, \widehat{(0,j)})$ this is (R4); dually (L, Q_0), $(Q_0, [0, i])$ and $(Q_0, \widehat{[0,j]})$ give (R5). The edges $(P_0, \widehat{[0,a]})$, $(L_0, \widehat{(0,a)})$, $(\widehat{(0,0)}, \widehat{[0,1]})$, $(\widehat{(0,1)}, \widehat{[0,0]})$ are the links, and the deleted edge $((0, a), [0, a])$ is (R1). The flag (P, L) plays no role in the construction: P is one of the points of ℓ^* and L one of the lines of Λ^*.

1.3. **What the new elements are: a crossed doubling of the frame.** The hat point H_s is a copy of X_s that does not see ℓ; the hat line K_s is a copy of λ_s that does not see A; and (Q, M) is a copy of the anti-flag (A, ℓ) that does not see the elements X_t, λ_t, $t \in T$. The cleanest way to say this is as a doubling of the frame, in the sense of Delorme [3], with one twist.

Let $\mathcal{F}$ be the set of frame elements A, ℓ, X_s, λ_s ($s \in S$), and give each $x \in \mathcal{F}$ a copy x' of the same kind:

$$A' = Q, \qquad \ell' = M, \qquad X'_s = H_s, \qquad \lambda'_s = K_s.$$

The *crossed doubling* $\mathcal{D}$ of the frame has the points and lines of Π and these copies, and the following incidences: all incidences of Π; for $x \in \mathcal{F}$ and $y \notin \mathcal{F}$ with $x \mathrel{\mathrm{I}} y$, the incidence $x' \mathrel{\mathrm{I}} y$ (a copy sees the same elements outside the frame as its original); for $x, y \in \mathcal{F}$ with $x \mathrel{\mathrm{I}} y$, the incidence $x' \mathrel{\mathrm{I}} y'$

three hats [1]

	ℓ	λ_0	λ_1	λ_a	M	K_0	K_1	K_a
A		•	•	•				•
X_0	•	•				•		
X_1	•		•				•	
X_a	•			○	•			•
Q				•		•	•	•
H_0		•			•		•	
H_1			•		•	•		
H_a	•			•	•			

four hats

	ℓ	λ_0	λ_c	λ_1	λ_a	M	K_0	K_c	K_1	K_a
A		•	•	•	•					•
X_0	•	○	•				•			
X_c	•		○	•				•		
X_1	•	•		○					•	
X_a	•				○	•				•
Q					•		•	•	•	•
H_0		•				•		•		
H_c			•			•			•	
H_1				•		•	•			
H_a	•				•	•				

TABLE 2. The *gadget*: all incidences among the frame and the new elements (• incident, ○ a flag of Π that was deleted). Every other incidence of a new element is with an element outside the frame, by rules (R2)–(R5). In both tables the block $\{X_t\} \times \{\lambda_t\}$ ($t \in T$) is the permutation matrix of ρ and the block $\{H_t\} \times \{K_t\}$ is that of π: the identity and the transposition (0 1) for the three hats, the 3-cycle σ twice for the four hats.

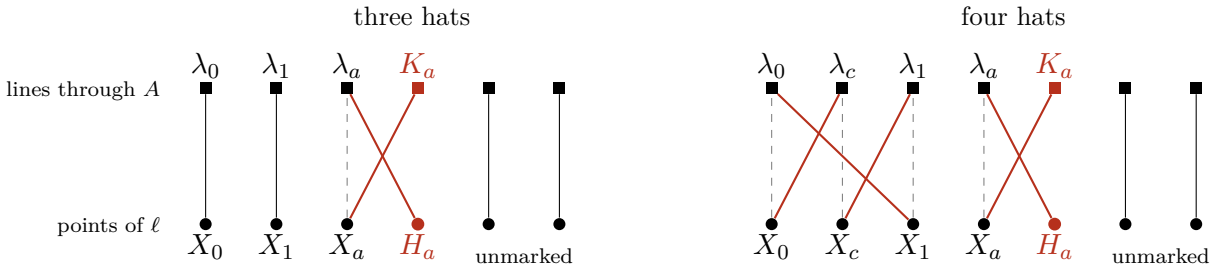


FIGURE 1. The incidences between the points of ℓ and the lines through A (squares are lines, circles points; black: kept flags of Π, dashed: deleted flags, red: new). In Π they form a perfect matching, $X \leftrightarrow AX$. In $\mathcal{S}$ they still form a perfect matching, once the new point H_a (which lies on ℓ) is counted among the points of ℓ and the new line K_a (which passes through A) among the lines through A: X_a is now matched with K_a and λ_a with H_a. The three hats keep the other flags. The four hats replace the three flags (X_t, λ_t), $t \in T$, by the cyclic shift $(X_t, \lambda_{\sigma(t)})$, the *rotation*. There are no other incidences between these points and lines.

(copies see copies), except that each flag (X_s, λ_s) is *crossed*: instead of $X'_s \mathrel{\mathrm{I}} \lambda'_s$ we put $X'_s \mathrel{\mathrm{I}} \lambda_s$ and $X_s \mathrel{\mathrm{I}} \lambda'_s$.

Reading off the neighbours of the copies: H_s lies on M (as $X_s \mathrel{\mathrm{I}} \ell$) and on the lines through X_s other than ℓ (those other than λ_s are outside $\mathcal{F}$; λ_s by the crossing); K_s contains Q and the points of λ_s other than A; M contains the hat points and the unmarked points of ℓ; Q lies on the hat lines and the lines through A other than the λ_s. Comparing with Definition 1.1:

Proposition 1.2. *$\mathcal{S}$ is obtained from the crossed doubling $\mathcal{D}$ of the frame by*

(i) *adding the* patches $X_a \mathrel{\mathrm{I}} M$, $Q \mathrel{\mathrm{I}} \lambda_a$, $A \mathrel{\mathrm{I}} K_a$, $H_a \mathrel{\mathrm{I}} \ell$ *and* $H_t \mathrel{\mathrm{I}} K_{\pi(t)}$ $(t \in T)$*;*
(ii) *deleting the flags* (X_s, λ_s), $s \in S$, *and adding the flags* $(X_t, \lambda_{\rho(t)})$, $t \in T$ *(for the three hats: deleting* (X_a, λ_a)*).*

So the question "what happens to the combinatorics when these points and lines are added" splits into two: what the doubling does (Lemma 1.3), and what the patches, the deletions and the rotation do (§1.4).

Lemma 1.3. *Suppose ℓ has an unmarked point ($|S| \le q$). In the crossed doubling $\mathcal{D}$:*

(a) *X_s and λ_s have degree $q+2$; A, ℓ, Q, M, H_s, K_s have degree $q+1$; an element outside $\mathcal{F}$ has degree $q+2$ if it is on ℓ or on some λ_s (a point), or passes through A or some X_s (a line), and degree $q+1$ otherwise.*

(b) *The pairs of points that are not collinear are exactly*

$$\{H_s, X_t\}\ (s \neq t) \qquad \text{and} \qquad \{H_s, Q\},$$

and the pairs of lines that are not concurrent are exactly $\{K_s, \lambda_t\}$ $(s \neq t)$ and $\{K_s, M\}$.

Proof. (a) X_s gains K_s (crossing) and λ_s gains H_s; A and ℓ gain nothing, since their frame neighbours are copied together with them. Q lies on $|S|$ hat lines and $q+1-|S|$ lines through A; H_s on M and q lines; dually for M, K_s. A point outside $\mathcal{F}$ gains M if it is on ℓ and K_s if it is on λ_s (at most one of these, as it is not A or an X_s).

(b) Two points of Π keep their common line. Q and a point u of Π: if $u = A$, use a line through A outside $\mathcal{F}$ (there is one, as $|S| \le q$; for the four hats with $q = 3$ there is none, and $\{Q, A\}$, $\{M, \ell\}$ are two more far pairs, repaired by the patches Q I λ_a and X_a I M); if Au is not a λ_s, use Au; if $u \neq A$ is on λ_s, use K_s. H_s and a point $u \notin \ell$: use uX_s, which is not ℓ. H_s and X_s: a line through X_s outside $\mathcal{F}$. H_s and an unmarked point of ℓ: M. Two hat points: M. This leaves H_s and X_t $(t \neq s)$, whose only common line in Π is ℓ, which H_s is not on, while $M \not\ni X_t$ and the other lines of H_s meet ℓ only in X_s; and H_s and Q, which have no common line (Q is on no line through X_s except possibly λ_s, and $Q \notin \lambda_s$; $Q \notin M$). The statement for lines is the same argument with points and lines exchanged; $\mathcal{D}$ is built from Π and the frame by rules that are symmetric under this exchange. □

So the copies cost nothing among the old points: no two points of Π become non-collinear and no two lines of Π stop meeting. The failures all involve a copy, and they come from paths of length 2 inside the frame: $X_s - \ell - X_t$ (the copy H_s of X_s does not see ℓ, and the original X_t does not see the copy M of ℓ) and $X_s - \lambda_s - A$ (the crossing removes the copy $H_s - K_s - Q$ of this path). For Delorme's doubling without crossing, the hat lemma (`POLISHED.md`, Lemma 1.3) says that a set of elements of a projective plane can be doubled without changing the diameter only if each of its connected components is a single element or a flag; the frame contains the paths ℓ, X_s, λ_s, A, so some patching is unavoidable. Lemma 1.3(a) also says that the doubling leaves exactly one free incidence at each of the $2|S|+4$ elements A, ℓ, Q, M, H_s, K_s, and none at X_s, λ_s.

1.4. **The patches, the deletions and the rotation.** The patches of Proposition 1.2(i) spend the free incidences of A, ℓ, Q, M, H_s, K_s, one each. The far pairs of Lemma 1.3(b) are repaired as follows (a pair $\{H_s, X_t\}$ always has $s \neq t$):

far pair in $\mathcal{D}$	common line in $\mathcal{S}$	because of
$\{H_s, X_a\}$	M	patch X_a I M
$\{H_a, X_t\}$	ℓ	patch H_a I ℓ
$\{H_t, X_{\pi(t)}\}$, $t \in T$	$K_{\pi(t)}$	link H_t I $K_{\pi(t)}$ (and $X_{\pi(t)}$ I $K_{\pi(t)}$)
$\{H_t, X_{\sigma^{-1}(t)}\}$, $t \in T$ (four hats)	λ_t	rotation $X_{\sigma^{-1}(t)}$ I λ_t (and H_t I λ_t)
$\{H_a, Q\}$	λ_a	patch Q I λ_a (and H_a I λ_a)
$\{H_t, Q\}$, $t \in T$	$K_{\pi(t)}$	link H_t I $K_{\pi(t)}$

The far pairs of lines are repaired dually: $\{K_s, \lambda_a\}$ by Q, $\{K_a, \lambda_t\}$ by A, $\{K_{\pi(t)}, \lambda_t\}$ by H_t, $\{K_t, \lambda_{\sigma(t)}\}$ by X_t (four hats), $\{K_a, M\}$ by X_a, $\{K_{\pi(t)}, M\}$ by H_t.

Why the three hats need no rotation and the four hats do. The pairs $\{H_s, X_t\}$ with $s = a$ or $t = a$ are repaired by ℓ and M. What remains is $\{H_t, X_u\}$ for $t \neq u$ in T: each hat point H_t must reach the $|T| - 1$ points X_u, $u \in T \setminus \{t\}$, and it has one free incidence, which goes to a link. For the three hats $|T| - 1 = 1$ and the link H_t I $K_{\pi(t)}$ suffices. For the four hats $|T| - 1 = 2$: the link reaches $X_{\sigma(t)}$, and $X_{\sigma^{-1}(t)}$ is reached through λ_t, which already carries H_t, after the rotation has moved $X_{\sigma^{-1}(t)}$ onto λ_t. Hence $\rho = \pi = \sigma$: $\rho^{-1}(t)$ must be the element of $T \setminus \{t\}$ that π does not reach. (Dually each hat line K_t meets $\lambda_{\sigma(t)}$ in X_t and $\lambda_{\sigma^{-1}(t)}$ in $H_{\sigma^{-1}(t)}$.) With $|T| = 4$ the same count gives a hat point two ways to reach three points, which is one heuristic reason why the pattern stops at four hats; a SAT search confirms that no fifth hat can be added inside this family for $q = 4, 5, 7, 8, 9, 11, 13$ (`research/plus2/REPORT.md`).

Why the flags (X_s, λ_s) are deleted. The patches X_a I M and Q I λ_a raise the degrees of X_a and λ_a, which were already $q + 2$, to $q + 3$; deleting the flag (X_a, λ_a) brings both back to $q + 2$. The rotation is degree-neutral: X_t trades λ_t for $\lambda_{\sigma(t)}$, and λ_t trades X_t for $X_{\sigma^{-1}(t)}$. Deleting a flag (X, λ) from a projective plane makes exactly q pairs of points non-collinear (the pairs $\{X, Y\}$, Y a point of λ other than X) and q pairs of lines non-concurrent (the pairs $\{\lambda, N\}$, N a line through X other than λ). After the deletions and the rotation, the pairs of points of Π that have no common line of Π any more are:

- three hats: $\{X_a, Y\}$ for the q points $Y \neq X_a$ of λ_a, and dually $\{\lambda_a, N\}$ for the q lines $N \neq \lambda_a$ through X_a;
- four hats: $\{X_s, Y\}$ for $s \in S$ and the $q - 1$ points Y of λ_s other than A and X_s, and $\{X_a, A\}$; dually $\{\lambda_s, N\}$ for the $q - 1$ lines N through X_s other than ℓ and λ_s, and $\{\lambda_a, \ell\}$. That is $4(q - 1) + 1$ pairs of each kind. The pairs $\{X_t, A\}$ and $\{\lambda_t, \ell\}$, $t \in T$, are not broken, because X_t and A are both on $\lambda_{\sigma(t)}$, and λ_t and ℓ both contain $X_{\sigma^{-1}(t)}$.

Each broken pair is repaired by the crossing and one patch: X_s and the points of λ_s other than A are all on K_s (rule (R3)), and A I K_a repairs $\{X_a, A\}$; dually λ_s and the lines through X_s other than ℓ all pass through H_s, and H_a I ℓ repairs $\{\lambda_a, \ell\}$. This is why the flags may be deleted at all: the crossing has already given every point of λ_s a second line through X_s (namely K_s), and every line through X_s a second point on λ_s (namely H_s).

Figure 2 shows the local picture at a hat pair; Table 2 shows the whole gadget.

1.5. **Degrees and counts.**

Lemma 1.4. *In $\mathcal{S}$ every element has degree $q+1$ or $q+2$. The points of degree $q+1$ (the* deficient *points) are the points of Π off ℓ, other than A, on the lines joining A to the unmarked points of ℓ; dually, the lines of degree $q+1$ are the lines of Π not through A, other than ℓ, through the unmarked points of ℓ. There are $(q-1)(q+1-|S|)$ of each: $(q-1)(q-2)$ for the three hats and $(q-1)(q-3)$ for the four hats. Each deficient point lies on exactly $q - |S|$ deficient lines, and each deficient line has exactly $q - |S|$ deficient points.*

Proof. Count the lines through each point: those of Π, minus deleted flags, plus added ones. A: $q+1$ lines and K_a. An unmarked point of ℓ: $q+1$ lines and M. X_a: loses λ_a, gains M and K_a. X_t, $t \in T$: four hats, loses λ_t and gains $\lambda_{\sigma(t)}$ and K_t; three hats, gains K_t. A point $p \notin \ell$, $p \neq A$: it is on exactly one line through A, the line Ap; it loses nothing, and gains K_s exactly when $Ap = \lambda_s$. Q: $|\Lambda^*| + |S| = (q + 1 - |T|) + |S| = q + 2$. H_s: q lines by (R2), M, and one link line. The lines are the dual count (Lemma 1.6 below). A deficient point p lies on Ap and on q lines meeting ℓ in the q points other than $Ap \cap \ell$; among these, $q - |S|$ are unmarked. □

So $\mathcal{S}$ has $q^2+q+1+|S|+1$ points and as many lines: $2(q^2+q+5)$ elements for the three hats and $2(q^2 + q + 6)$ for the four hats. To make it regular, join each deficient point to a deficient line not through it, by a perfect matching. Such a matching exists when there are deficient elements ($|S| \leq q$; for the four hats with $q = 3$ the structure is already regular): the bipartite graph of non-incident

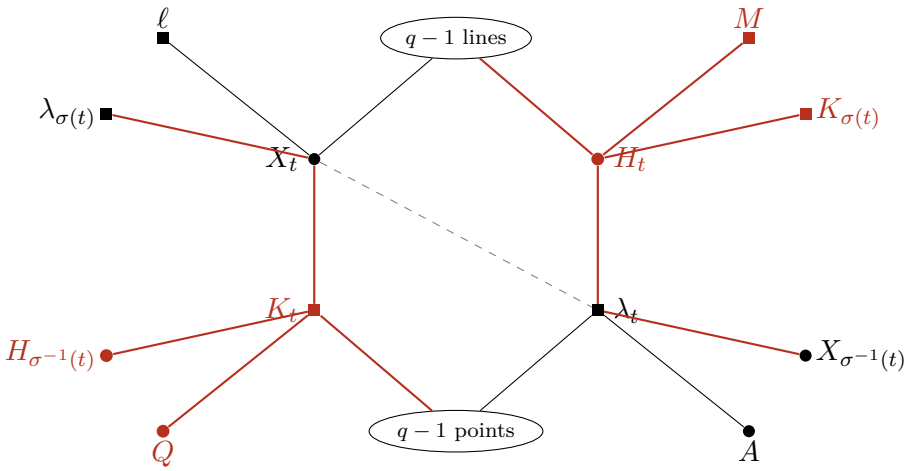


FIGURE 2. The local picture at a hat pair (H_t, K_t), $t \in T$, of the four hats, in the incidence graph. Circles are points, squares are lines, new elements and new incidences are red, the dashed edge is the deleted flag (X_t, λ_t). The upper blob is the set of $q-1$ lines through X_t other than ℓ and λ_t, the lower blob the $q-1$ points of λ_t other than A and X_t. H_t and X_t are twins on the upper blob, K_t and λ_t twins on the lower blob, and the flag (X_t, λ_t) is replaced by the crossed pair X_t I K_t, H_t I λ_t. Every element shown has degree $q+2$. For the three hats ($t \in \{0,1\}$) the dashed flag is kept, the edges to $\lambda_{\sigma(t)}$ and $X_{\sigma^{-1}(t)}$ are absent, and the links go to $K_{\pi(t)}$ and $H_{\pi(t)}$, $\pi = (0\ 1)$.

pairs between deficient points and deficient lines is regular of degree $(q-1)(q+1-|S|)-(q-|S|) > 0$, and a regular bipartite graph of positive degree has a perfect matching (König). Adding flags cannot increase distances. (In $\mathrm{PG}(2,q)$ explicit matchings are given in coordinates in [1] and in `coords.tex`; any matching of non-incident pairs works.)

1.6. **The diameter.**

Lemma 1.5 (Points)**.** *Any two distinct points of $\mathcal{S}$ are collinear in $\mathcal{S}$; Table 3 gives a common line for every pair.*

Proof. We check that Table 3 covers every pair and that each named line contains both points. The only flags of Π missing from $\mathcal{S}$ are among the (X_s, λ_s); so a flag (u, L) of Π survives whenever $u \notin \ell$ or L does not pass through A.

Two points of Π. If the line uv of Π lost an incidence, then (up to order) $u = X_s$, $uv = \lambda_s$ with the flag deleted, and v is A or another point of λ_s. If $v \neq A$, both lie on K_s by (R3). If $v = A$ and $s = a$: X_a I K_a by (R3) and A I K_a by (R6). If $v = A$ and $s = t \in T$ (four hats): the rotation puts X_t on $\lambda_{\sigma(t)}$, which passes through A.

Q and a point u of Π. $\lambda_a \in \Lambda^*$ passes through A and carries Q. K_s contains X_s and Q. An unmarked point u of ℓ: $Au \in \Lambda^*$, and the flag (u, Au) was not touched. A point $u \notin \ell$, $u \neq A$: the flag (u, Au) survives; if $Au \in \Lambda^*$ it carries Q; otherwise $Au = \lambda_t$, $t \in T$, and u is a point of λ_t other than A, so u I $K_t \ni Q$.

H_s and a point u of Π. If $u \notin \ell$, the line uX_s of Π is not ℓ, so it carries H_s, and it contains u in $\mathcal{S}$. If $u = X_s$, any of the $q-1 \geq 1$ lines through X_s other than ℓ and λ_s carries both. If $u \in \ell^*$ (this includes X_a when $s \neq a$), use M. The remaining case is $u = X_{t'}$ with $t' \in T \setminus \{s\}$. If $s = a$, then H_a I ℓ. If $s = t \in T$: for the three hats $t' = \pi(t)$; for the four hats $t' \in \{\sigma(t), \sigma^{-1}(t)\}$, as σ is

pair of points	common line	three hats
two points u, v of Π		
the line uv of Π keeps both incidences	uv	
$u = X_s$, v a point of λ_s other than A, X_s, flag (X_s, λ_s) deleted	K_s	only $s = a$
$u = X_a$, $v = A$	K_a	
$u = X_t$, $v = A$, $t \in T$ (four hats)	$\lambda_{\sigma(t)}$	λ_t is kept
Q and a point u of Π		
$u = A$	λ_a	
$u = X_s$	K_s	
u an unmarked point of ℓ	Au	
$u \notin \ell$, $u \neq A$, $Au \in \Lambda^*$	Au	
$u \notin \ell$, $u \neq A$, $Au = \lambda_t$ with $t \in T$	K_t	
H_s and a point u of Π		
$u \notin \ell$ (including $u = A$)	uX_s	
$u = X_s$	a line through X_s other than ℓ, λ_s	
$u \in \ell^*$, $u \neq X_s$	M	
$s = a$, $u = X_t$ with $t \in T$	ℓ	
$s = t \in T$, $u = X_{\pi(t)}$	$K_{\pi(t)}$	$\pi = (0\ 1)$
$s = t \in T$, $u = X_{\sigma^{-1}(t)}$ (four hats)	λ_t	does not occur
two new points		
H_s and $H_{s'}$	M	
Q and H_a	λ_a	
Q and H_t, $t \in T$	$K_{\pi(t)}$	

TABLE 3. A common line for every pair of distinct points of $\mathcal{S}$. Here $s, s' \in S$ and $t \in T$. The third column notes where the three hats differ; elsewhere the entry is the same for both constructions.

a 3-cycle. The line $K_{\pi(t)}$ contains $X_{\pi(t)}$ by (R3) and H_t by the link. The line λ_t contains $X_{\sigma^{-1}(t)}$ by the rotation and H_t by (R2).

Two new points. M carries all hat points. λ_a carries Q (as $\lambda_a \in \Lambda^*$) and H_a (as a line through X_a other than ℓ). $K_{\pi(t)}$ carries Q and, by the link, H_t. □

For lines we use duality. In the dual plane Π^D the line ℓ is a point and A a line not through it, the lines λ_s are points on the line A, and the line joining the points ℓ and λ_s is the point X_s.

Lemma 1.6 (Duality)**.** *The dual of $\mathcal{S}(\Pi; A, \ell, (X_s), \pi, \rho)$ is $\mathcal{S}(\Pi^D; \ell, A, (\lambda_s), \pi^{-1}, \rho^{-1})$, by the identity on the elements of Π and $H_s \leftrightarrow K_s$, $Q \leftrightarrow M$. Consequently any two distinct lines of $\mathcal{S}$ are concurrent.*

Proof. Read the six rules in Π^D with the frame $(\ell, A, (\lambda_s))$. Rule (R1) deletes the flags (λ_s, X_s) and adds λ_t I $X_{\rho^{-1}(t)}$, which is the rotation X_u I $\lambda_{\rho(u)}$ of $\mathcal{S}$. Rule (R2) gives a new element incident with every element incident with λ_s other than A: this is K_s; rule (R3) gives one incident with every element incident with X_s other than ℓ: this is H_s. Rule (R4) gives a new element incident with the lines through A other than the λ_t, $t \in T$, and with all K_s: this is Q; likewise (R5) gives M. Rule (R6) gives ℓ I H_a, K_a I A and K_t I $H_{\pi^{-1}(t)}$, which are the links of $\mathcal{S}$. Since Π^D is a projective plane of order q and π^{-1}, ρ^{-1} are again of the required kind (for the three hats $\pi^{-1} = \pi$,

$\rho^{-1} = \rho$; for the four hats $\pi^{-1} = \rho^{-1} = \sigma^{-1}$, a 3-cycle), Lemma 1.5 for the dual structure says that any two lines of $\mathcal{S}$ are concurrent. □

Theorem 1.7. *Let Π be a projective plane of order $q \geq 3$. The three-hat structure of* [1] *and the four-hat structure are incidence structures with all degrees $q+1$ or $q+2$ and diameter 3, with $2(q^2+q+5)$ and $2(q^2+q+6)$ elements. Adding a perfect matching of non-incident pairs between the deficient points and lines makes them $(q+2)$-regular. Hence $n_b(q+2, 3) \geq 2(q^2+q+6)$ for every prime power $q \geq 3$.*

Proof. Degrees and counts are Lemma 1.4. Every element is incident with something. By Lemmas 1.5 and 1.6 any two points are collinear and any two lines concurrent, so by Lemma 0.1(a) the diameter is at most 3; it is 3 because $A \not\mathrel{I} \ell$. □

Only the axioms of a projective plane are used, so the theorem holds in non-Desarguesian planes too (we checked the nearfield plane of order 9). In the range of the tables, $q = 9, 11, 13$ give 192, 276, 376 at $\Delta = 11, 13, 15$, two more than [1].

1.7. **From three hats to four, step by step.** Start from the three hats with labels $0, 1, a$ and choose a fourth point X_c of ℓ, different from X_0, X_1, X_a. The four-hat structure arises by exactly these changes:

(1) add the point H_c, on the q lines through X_c other than ℓ, on M and on K_1; add the line K_c, through the q points of λ_c other than A, through Q and through H_0;
(2) delete the flags (X_0, λ_0), (X_c, λ_c), (X_1, λ_1), and add (X_0, λ_c), (X_c, λ_1), (X_1, λ_0);
(3) delete $X_c \mathrel{I} M$ and $Q \mathrel{I} \lambda_c$ (now X_c is a T-point and λ_c a T-line);
(4) delete the link $H_0 \mathrel{I} K_1$; it is replaced by $H_0 \mathrel{I} K_c$ and $H_c \mathrel{I} K_1$ from (1), while $H_1 \mathrel{I} K_0$ is kept.

In terms of the links, the cycle $0 \to 1 \to 0$ of $\pi = (0\ 1)$ becomes the cycle $0 \to c \to 1 \to 0$ of σ. The four-hat structure has $2q+1$ more flags than the three-hat structure: $2(q+2)$ at H_c and K_c, three from the rotation, minus six. (Checked by computer against the coordinate versions for $q = 4, 5, 7$ and every choice of a, c.)

2. Diameter 4: clones and shadow lines in generalized quadrangles

A *generalized quadrangle* of order (s, t) is an incidence structure in which every line has $s+1$ points, every point is on $t+1$ lines, and

for every anti-flag (p, L) there is *exactly one* point of L collinear with p.

It has $(s+1)(st+1)$ points and $(t+1)(st+1)$ lines, and its incidence graph has diameter 4 and girth 8. By Lemma 0.1(b), an incidence structure has diameter at most 4 as soon as for every anti-flag (p, L) there is *at least one* point of L collinear with p. All constructions in this section only add points, lines and incidences to a quadrangle; nothing is deleted or moved, so no pair of points stops being collinear.

This is the first difference with diameter 3. In a projective plane two cloned points (new points on all lines through an old point) would overload the line joining them, and the hats of Section 1 are clones of the points X_s that avoid the common line ℓ. In a quadrangle, a clone of a point inherits the covering property from its point at once; what has to be arranged is only that every line receives the same number of clones, and this is what an m-ovoid is.

2.1. **Clones and shadow lines.** Let $\mathcal{S}$ be a generalized quadrangle of order (s, t).

- A *clone* of a point p is a new point incident with exactly the lines of $\mathcal{S}$ through p.
- A *shadow line* over a line L is a new line that contains, for each point p of L, exactly one point *over* p: either p itself or one of the clones of p, called the *stand-in* for p.

Definition 2.1 (Theorem B of the lab)**.** Let w assign to every point p a number $w(p) \geq 0$, and let $\mathcal{M}$ be a finite list of lines of $\mathcal{S}$ (repetitions allowed). The structure $\mathcal{S}[w, \mathcal{M}]$ has the points and lines of $\mathcal{S}$, $w(p)$ clones of each point p, and one shadow line over each entry of $\mathcal{M}$, with a chosen stand-in for each of its points. Its incidences are those of $\mathcal{S}$, those of the clones, and those of the shadow lines with their points.

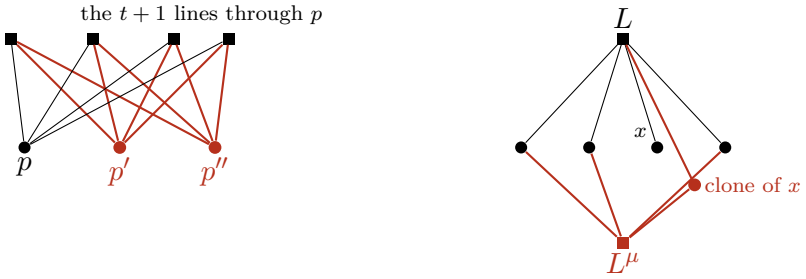


FIGURE 3. Left: a point p with two clones p', p''; all three are on the $t+1$ lines through p and on no other line of $\mathcal{S}$. Right: a shadow line L^μ over L; for the point x of L the stand-in is a clone of x, for the other points of L it is the point itself. The clone of x is on L as well, being on every line through x.

What happens to the combinatorics. No incidence is deleted, so every pair of old points that was collinear stays collinear. A point p and its clones are pairwise collinear, on $t+1$ common lines; two lines through p now meet in $1+w(p)$ points; a shadow line whose stand-ins are all original has the same points as its line. So the structure is no longer a partial linear space, but these coincidences are harmless. Two old points that were not collinear stay non-collinear: they have no common old line, a shadow line over L contains only points over points of L, and an old point is over itself only.

Why the diameter is 4. Write $\bar{x}$ for the old point under a point x ($\bar{x} = x$ for an old point, $\bar{x} = p$ for a clone of p) and $\bar{Y}$ for the old line under a line Y. Every point x is on every old line through $\bar{x}$. Let (x, Y) be an anti-flag.

- Y old. Then $\bar{x} \not\mathrm{I} Y$; let r be the point of Y collinear with $\bar{x}$, on the line N. Then $x \mathrm{I} N \mathrm{I} r \mathrm{I} Y$.
- Y a shadow line, $\bar{x} \not\mathrm{I} \bar{Y}$. Let r be the point of $\bar{Y}$ collinear with $\bar{x}$, on the line N, and r^* the stand-in for r on Y. Then $x \mathrm{I} N \mathrm{I} r^* \mathrm{I} Y$, as r^* is on every old line through r.
- Y a shadow line, $\bar{x} \mathrm{I} \bar{Y}$. Take a point $r \neq \bar{x}$ of $\bar{Y}$ and its stand-in r^*: then $x \mathrm{I} \bar{Y} \mathrm{I} r^* \mathrm{I} Y$.

So in every case some point of Y is collinear with x, and the diameter is at most 4 (Lemma 0.1(b)); it is 4 because non-collinear old points remain.

Degrees. An old line L has $s + 1 + \sum_{p \mathrm{I} L} w(p)$ points. A shadow line has $s+1$ points. An old point or a clone is on $t+1$ old lines plus the shadow lines for which it is the stand-in. This proves:

Theorem 2.2 (Theorem B)**.** *For every generalized quadrangle $\mathcal{S}$ of order (s,t), every w, $\mathcal{M}$ and choice of stand-ins, $\mathcal{S}[w, \mathcal{M}]$ has diameter* 4*, $(s+1)(st+1)+\sum_p w(p)$ points and $(t+1)(st+1)+|\mathcal{M}|$ lines. Its maximum degree is at most Δ if and only if $\Delta \geq \max(s,t)+1$, every line of $\mathcal{S}$ carries clones of total number at most $\Delta - s - 1$, and every point (old or clone) is the stand-in for at most $\Delta - t - 1$ entries of $\mathcal{M}$.*

2.2. **Exact weights: m-ovoids and covers.** A *weighted m-ovoid* is a weight w on the points such that every line carries total weight exactly m; equivalently, a multiset of points meeting every line in exactly m points, counted with multiplicity. It is an *m-ovoid* if it is a set, and an *ovoid* if moreover $m = 1$ (a set of points meeting every line once). Dually, a *weighted m-cover* is a multiset of lines with exactly m members through every point, and a *spread* is a set of lines partitioning the points.

Corollary 2.3. *Let $\Delta \geq \max(s,t)+1$, let w be a weighted $(\Delta-s-1)$-ovoid and $\mathcal{M}$ a weighted $(\Delta-t-1)$-cover, with every stand-in original. Then every old element of $\mathcal{S}[w,\mathcal{M}]$ has degree exactly Δ, every clone $t+1$ and every shadow line $s+1$, and there are $2\Delta(st+1)$ elements in all. Joining the clones to the shadow lines by a suitable set of anti-flags makes the structure Δ-regular, with diameter still 4.*

(Counting flags gives $\sum_p w(p) = (\Delta-s-1)(st+1)$ and $|\mathcal{M}| = (\Delta-t-1)(st+1)$. No clone is on a shadow line, so all clone–shadow pairs are anti-flags, and the missing incidences can be added in any balanced way, for instance cyclically; `POLISHED.md`, Corollary 3.3.)

So for diameter 4 the whole question is: *which quadrangles carry weighted m-ovoids and covers for the right m?* Three answers give the lab's values.

2.3. **Theorem A: two elliptic quadrics in the Ahrens–Szekeres quadrangle.** Let q be odd and $\Delta = q+2$. The quadrangle is $\mathrm{AS}(q)$, of order $(q-1,q+1)$; then $\Delta-s-1=2$ and $\Delta-t-1=0$, so we need a *weighted 2-ovoid of* $\mathrm{AS}(q)$ *and no shadow lines*. Every point of $\mathrm{AS}(q)$ is already on $t+1=q+2$ lines, so the clones have full degree and the result is $(q+2)$-regular.

The quadrangle. $\mathrm{AS}(q)$ is Payne's derived quadrangle of the symplectic quadrangle $W(q)$ at a point x [4, 5], and its definition is combinatorial. For points u,v write $\{u,v\}^{\perp}$ for the set of points collinear with both, and $\{u,v\}^{\perp\perp}$ for the set of points collinear with every point of $\{u,v\}^{\perp}$; in $W(q)$, for u,v not collinear, $\{u,v\}^{\perp\perp}$ has $q+1$ points (every point of $W(q)$ is regular). Then:

- the points of $\mathrm{AS}(q)$ are the q^3 points of $W(q)$ not collinear with x (and different from x);
- its lines are (i) the lines of $W(q)$ not through x, each without its unique point collinear with x, and (ii) the sets $\{x,y\}^{\perp\perp}\setminus\{x\}$, y not collinear with x (the *hyperbolic lines*).

Every line has q points, every point is on $q+2$ lines, and the q^2 hyperbolic lines form a spread of $\mathrm{AS}(q)$.

The configuration used. Two sets $\mathcal{E}_1,\mathcal{E}_2$ of q^2 points of $\mathrm{AS}(q)$ such that

(E1) each hyperbolic line meets $\mathcal{E}_1$ in exactly one point and $\mathcal{E}_2$ in exactly one point;
(E2) every other line of $\mathrm{AS}(q)$ meets $\mathcal{E}_1$ and $\mathcal{E}_2$ in exactly two points in total (a point of $\mathcal{E}_1\cap\mathcal{E}_2$ counted twice).

Then $w = \mathbf{1}_{\mathcal{E}_1}+\mathbf{1}_{\mathcal{E}_2}$ is a weighted 2-ovoid. Such a pair exists for every odd q: two elliptic quadrics of $\mathrm{PG}(3,q)$ through x, both tangent to the plane of points collinear with x, in a suitable relative position; they meet in x and in one point e of $\mathrm{AS}(q)$ (`POLISHED.md`, Theorem 4.1; the proof is a discriminant computation, and a line of type (i) meets the two quadrics in $1+\chi$ and $1-\chi$ points for some $\chi\in\{-1,0,1\}$). So:

Clone every point of $\mathcal{E}_1$ and every point of $\mathcal{E}_2$ once (so e gets two clones).

Each line of $\mathrm{AS}(q)$ gains exactly two clones, each clone is on the $q+2$ lines of its point, nothing else changes, and Theorem 2.2 gives a $(q+2)$-regular structure of diameter 4 with $q^3+q^2(q+2)+2q^2 = 2q^2(q+2)$ elements. At $q=9,11,13$: 1782, 3146, 5070. Weights are needed: $\mathrm{AS}(3)$ has no 2-ovoid and $\mathrm{AS}(5)$ no m-ovoid for $m\leq 4$ (integer programs, `POLISHED.md` §4). And nothing more can be added at $\Delta=q+2$: points are full, and every line has room for exactly two clones.

2.4. **Proposition C: a 2-ovoid and a spread of $W(q)$.** Let q be odd and $\Delta=q+3$ in $W(q)$, of order (q,q); now $\Delta-s-1=\Delta-t-1=2$. The configuration used is a 2-*ovoid* $\mathcal{O}$ of $W(q)$ (a set of $2(q^2+1)$ points meeting every line in exactly two points) and a *spread* Σ (q^2+1 lines partitioning the points). The construction:

Clone every point of $\mathcal{O}$ once, and put two shadow lines over every line of Σ.

Every old line gains two clones, every old point is on two shadow lines (those over its spread line), clones and shadow lines have degree $q+1$, and Corollary 2.3 completes it to a $(q+3)$-regular structure of diameter 4 with $2(q^2+1)(q+3)$ elements; one explicit completion joins the i-th clone to the shadow lines $2i$ and $2i+1$ (indices modulo $2(q^2+1)$, in any numbering). At $q=13$ this is 5440

at $\Delta = 16$. 2-ovoids of $W(q)$, q odd, exist: the union of two suitable classes of Ebert's partition of $\mathrm{PG}(3, q)$ into elliptic quadrics (orbits of a Singer subgroup) is one (`POLISHED.md`, Proposition 5.1); see also [2]. Spreads of $W(q)$ exist for every q.

2.5. **Even q, and all degrees.** For even q, $W(q)$ has both an ovoid $\mathcal{O}$ and a spread Σ [5]. With j clones of every point of $\mathcal{O}$ and j shadow lines over every line of Σ, Corollary 2.3 gives a $(q+1+j)$-regular structure of diameter 4 with $2(q+1+j)(q^2+1)$ elements. Combining the three cases (ovoid and spread for even q; $j/2$ copies of a 2-ovoid and j copies of a spread for odd q and even j; $\mathrm{AS}(q)$ with $(\Delta-q)/2$ copies of the weighted 2-ovoid of Theorem A and $\Delta-q-2$ copies of the hyperbolic spread for odd q and odd j) gives $n_b(\Delta, 4) \geq 2q^2\Delta$ for every prime power $q \leq \Delta - 1$ (`POLISHED.md`, Corollary 5.3).

2.6. **For comparison: Delorme's doubling, and stand-ins.** *Delorme's construction* behind the 2013 values [3] takes a spread Σ of $W(q)$ and a line $v \in \Sigma$. Each line of Σ other than v gets one copy with the same points (a shadow line with all stand-ins original); each point of v gets one copy, which is on the q lines through the point other than v; and v gets three copies, one through the original points of v and two through the copies. All degrees are at most $q+2$, the diameter is 4, and there are $2(q+1)(q^2+1)+q^2+q+4$ elements. It is a doubling in Delorme's sense: the copied set is the spread with the points of one of its lines, at two levels. Theorem 2.2 is not a doubling: since a clone is on *every* old line through its point, no condition on the shape of the copied set is needed, and whole m-ovoids can be cloned.

Stand-ins: the $(7, 4)$ *example.* In the quadrangle $Q(4, 5)$ (order $(5, 5)$) take an ovoid $\mathcal{O}$ and clone each of its 26 points once: every line gains one clone and has degree 7, every old point has degree 6. Shadow lines can now be added as long as every point serves at most one of them: a point off $\mathcal{O}$ has no clone and can be on at most one shadow line, while a point of $\mathcal{O}$ can be on two, standing in for itself on one and letting its clone stand in on the other. An integer program finds 21 such shadow lines (four points of $\mathcal{O}$ are on two of them), giving $2 \cdot 6 \cdot 26 + 26 + 21 = 359$ elements, maximum degree 7 and diameter 4 (`POLISHED.md`, Appendix A). In this family at most $q^2 - q + 2$ shadow lines are possible (`POLISHED.md`, Remark 3.5).

3. Diameter 6

To be written.

4. Diameter 7

To be written.

Appendix A. Coordinates and checks

Explicit coordinates for every construction: `POLISHED.md` (§2 for the hats, §§3–5 for diameter 4, §§6–7 for diameter 6) and `D7_HEXAGON.md` (diameter 7). The check script for Section 1 is `research/synthetic/checks/check_d3.py`; its output is in `check_d3.out`.

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